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Full length article



An experimental information-theoretic framework for cognitive semantic communication: A rate–distortion–relevance study over RIS-enabled wireless links☆

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ABSTRACT

This paper introduces a simulation-based experimental framework for cognitive semantic communication over Reconfigurable Intelligent Surface (RIS)-enabled wireless links, developed under an extended rate-distortion-relevance (RDR) model. Unlike traditional systems that prioritize bit-level fidelity, our framework emphasizes semantic preservation and energy efficiency under channel impairments. Based on simulation results, under challenging channel conditions at −10 dB SNR, the proposed approach maintains 65.9% accuracy compared to 54.5% for conventional Shannon-theoretic approach, while achieving 92.6% reduction in energy per semantic bit. All code and validation data are publicly available, providing a reproducible foundation for future 6G testbeds where semantic relevance is jointly optimized with physical-layer constraints.

1. Introduction

The vision for 6G wireless communication systems involves supporting massive data flows, ultra-reliable low-latency communication, and seamless integration of intelligent services [1]. Traditional Shannon-centric frameworks are fundamentally limited in such settings, as they focus on bit-level fidelity without accounting for the semantic meaning or relevance of the transmitted content [2,3].

Cognitive Semantic Communication (CSC) offers a paradigm shift by emphasizing the transmission of semantically significant information, rather than raw bits [4]. Inspired by human cognition and recent advances in neural networks, CSC systems can extract, prioritize, and interpret semantic content in context-aware scenarios. While traditional systems primarily measure performance through bit error rate (BER), semantic systems must balance three interconnected dimensions: rate (bandwidth efficiency), distortion (bit-level accuracy), and relevance (semantic meaningfulness); presenting a complex multi-objective optimization challenge.

In parallel, Reconfigurable Intelligent Surfaces (RIS) have emerged as a transformative technology for next-generation wireless systems, offering unprecedented performance enhancement capabilities through programmable manipulation of the wireless propagation environment [5]. RIS technology provides significant performance improvements across multiple dimensions: beamforming capabilities that enable precise signal direction control, coverage extension in challenging propagation environments, energy efficiency benefits through passive signal enhancement, and interference management through intelligent phase control of reflecting elements. These capabilities make RIS particularly suitable for 6G communications where diverse service requirements demand adaptive and efficient wireless infrastructure.

Recent advances in RIS technology have introduced Simultaneously Transmitting and Reflecting RIS (STAR-RIS), which extends the capabilities of conventional reflecting-only surfaces to achieve full-space 360-degree coverage [6,7]. Unlike traditional RIS that can only reflect incident signals to one side, STAR-RIS simultaneously transmits and reflects signals to both sides of the surface, eliminating geographical constraints and enabling more flexible deployment strategies [7]. STAR-RIS operates through three distinct protocols: energy splitting (ES), mode switching (MS), and time switching (TS), each offering different trade-offs between transmission and reflection capabilities. This bidirectional functionality is particularly advantageous for semantic communication systems, where users with different semantic priorities may be distributed across diverse geographical locations, requiring adaptive coverage patterns that conventional RIS cannot provide.

The integration of Index Modulation Multiple Access (IMMA) with RIS technology presents another promising avenue for enhancing spectral and energy efficiency in 6G networks [8]. IMMA extends conventional non-orthogonal multiple access (NOMA) by conveying additional information through the indices of system building blocks; such as transmit antennas, subcarriers, or RIS elements, while maintaining compatibility with traditional signal constellations [8]. When

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applied to RIS-aided networks, IMMA enables the transmission of semantic information through both the phase configuration patterns of RIS elements and conventional modulation schemes [9]. This multi-dimensional information encoding is particularly valuable in semantic communication scenarios, where different semantic priorities can be mapped to different modulation domains, creating hierarchical information structures that align naturally with cognitive communication principles.

The synergy between RIS technology and semantic communication becomes particularly evident when considering energy efficiency constraints and channel robustness requirements. The passive nature of RIS elements aligns exceptionally well with the computational demands of semantic encoding and decoding processes [10]. While semantic communication systems typically require significant processing power for neural network-based encoding and decoding, the energy-efficient signal enhancement provided by RIS can improve channel conditions sufficiently to reduce the complexity requirements of semantic processing algorithms. This energy-semantic synergy creates opportunities for more sophisticated cognitive communication strategies within practical power budgets, enabling the deployment of advanced semantic processing in resource-constrained environments typical of 6G applications.

Recent research has also highlighted the importance of advanced wireless communication modeling techniques that account for finite blocklength effects and covert communication requirements in next-generation systems [11,12]. These considerations are particularly relevant for semantic communication systems where information packets may have variable lengths depending on semantic content complexity, and where maintaining communication security while preserving semantic fidelity presents unique challenges.

Despite their individual merits, the joint application of CSC and RIS remains underexplored; particularly in terms of their combined impact on information-theoretic performance. While several studies have examined either semantic communication [4] or RIS-enabled transmission [10] separately, there exists a critical knowledge gap regarding how RIS-enabled physical-layer control interacts with semantic-level encoding and decoding strategies. This gap becomes particularly significant in challenging propagation environments where semantic preservation is paramount.

In this work, we propose a simulation-based experimental framework that unifies CSC and RIS technologies within a coherent information theoretic context. Specifically, we extend the classical rate-distortion model by incorporating semantic relevance as a third dimension, resulting in a rate-distortion-relevance (RDR) framework. Unlike previous studies that treat semantic communication as a black-box neural architecture, we develop explicit mathematical relationships between RIS configurations, channel conditions, and semantic performance metrics, validated through Monte Carlo simulation analysis.

Through comprehensive Monte Carlo simulations with 300 trials per configuration, our framework provides statistically validated evidence of RIS benefits for semantic communication. Our experimental results demonstrate that RIS provides substantial performance improvements, with up to 11.5 percentage point gains in challenging low-SNR conditions. Specifically, our proposed semantic-RIS system achieves 65.9% accuracy at −10 dB SNR compared to only 54.5% for conventional approaches without RIS, representing a statistically significant improvement with large effect sizes.

Our framework provides several unique insights:

- We provide definitive simulation-based experimental evidence that RIS significantly enhances semantic communication performance, with the largest benefits occurring in challenging channel conditions where semantic preservation is most critical.
- We establish through rigorous baseline comparisons that RIS-assisted links maintain semantic integrity even when conventional bit-level metrics indicate significant degradation, particularly in low-SNR environments.
- We identify fundamental trade-offs between energy efficiency and semantic preservation, showing that RIS configurations can optimize both dimensions simultaneously under specific channel conditions.
- We demonstrate that the combined semantic processing and RIS enhancement provides superior robustness compared to either approach individually.

Unlike prior studies that focus exclusively on bit-level performance or end-to-end neural architectures, our framework offers a system-level perspective with both theoretical modeling and empirical validation. Key contributions of this paper are:

- A novel RDR-based theoretical framework for evaluating CSC over RIS-assisted links, with mathematical formulations that connect physical-layer parameters to semantic performance metrics.
- Comprehensive experimental validation with proper statistical rigor, including baseline comparisons between RIS and non-RIS configurations that definitively demonstrate RIS benefits for semantic communication.
- Statistical significance analysis with confidence intervals and effect size calculations, providing publication-quality evidence for all performance claims.
- An energy efficiency model based on the energy per semantic bit (EPSB), revealing optimal operating points for semantic communication systems under energy constraints.
- Complete methodology documentation addressing simulation procedures, dataset specifications, and experimental reproducibility requirements.

While our current implementation focuses on bit-level semantic metrics as a first step toward full semantic understanding, the framework establishes a foundation for future extensions to higher-level semantic representations. By bridging the gap between information theory and semantic communication, this work contributes to the theoretical underpinnings necessary for practical 6G semantic systems.

The remainder of the paper is organized as follows: Section 2 discusses relevant background on CSC, RIS, and semantic information theory. Section 3 details the system architecture and experimental design. Section 4 presents the RDR theoretical model and its mathematical formulation. Section 5 provides results and analysis across various performance metrics. Section 6 concludes the paper with insights and directions for future research.

2. Background and related work

2.1. Cognitive semantic communications

Semantic communication marks a fundamental departure from classical Shannon theory by prioritizing the transmission of meaning rather than exact symbol reconstruction. In this paradigm, the objective shifts from error-free bit delivery to the preservation of task-relevant information [13]. By identifying and encoding semantically significant content, such systems offer the potential to reduce bandwidth usage and improve robustness in noisy or resource-constrained environments.

Cognitive Semantic Communication (CSC) builds upon this foundation by incorporating context-aware, cognitively inspired mechanisms that mimic human-like information processing [14]. These systems leverage advances in deep learning to model contextual semantics and adapt transmission strategies accordingly. Recent work has demonstrated the effectiveness of CSC in various domains, including image transmission [15], natural language communication [14], and speech processing [16].

While these developments highlight the promise of CSC, most existing approaches focus primarily on end-to-end learning and overlook the role of underlying physical-layer dynamics, particularly under real-world channel impairments.

2.2. Reconfigurable intelligent surfaces

Reconfigurable Intelligent Surfaces (RIS) have emerged as a transformative technology for next-generation wireless systems by enabling programmable manipulation of the wireless propagation environment [10]. An RIS comprises a large array of passive reflecting elements whose phase, amplitude, or polarization response can be dynamically adjusted. These surfaces enable functionalities such as directional beamforming, interference cancellation, and coverage extension without the energy overhead associated with active relays.

The deployment of RIS in wireless networks offers several compelling advantages, including improved signal quality, spectral efficiency, and energy performance [17]. However, the practical integration of RIS introduces challenges such as channel estimation, phase quantization effects, hardware constraints, and control overhead. Despite growing interest, relatively little attention has been paid to how RIS influences higher-layer semantic metrics, or how RIS-configurable environments affect semantic communication robustness.

2.3. Information theory for semantic communications

Classical information theory, formalized through the rate-distortion framework, quantifies the trade-off between transmission rate and reconstruction fidelity. However, this formulation treats all transmitted bits as equally important and does not account for the semantic relevance of information in specific tasks or applications.

To address this limitation, recent work has proposed extensions to include a relevance dimension; resulting in the rate-distortion-relevance (RDR) framework [18]. This model introduces a third axis of analysis: semantic relevance, typically measured via mutual information between the transmitted and inferred semantic representations. The RDR theory enables a more holistic evaluation of communication systems, balancing bitrate, fidelity, and semantic utility.

Although RDR provides a solid theoretical foundation, its practical application remains limited. Few studies have explored how it interacts with real-world wireless constraints, such as RIS-enhanced channels or Doppler impairments. Moreover, empirical validations of RDR-based communication strategies in hardware-in-the-loop or realistic fading environments are scarce; representing a key gap this paper aims to address.

2.4. RIS benefits for semantic communication

RIS technology provides unique advantages for semantic communication through controllable propagation environments that can be dynamically optimized for information pattern preservation. Unlike conventional systems that treat all transmitted bits uniformly, RIS-enabled semantic communication can create adaptive channel conditions that prioritize the preservation of critical information structures. This capability becomes particularly valuable in challenging propagation environments where semantic content preservation is paramount, as RIS can compensate for channel impairments by intelligently directing signal energy toward maintaining essential information patterns.

2.4.1. Phase diversity for semantic channels

RIS enables creation of "semantic channels" that provide differential protection based on content importance. For semantic content $\mathbf{s} = [\mathbf{s}_{\text{critical}}, \mathbf{s}_{\text{important}}, \mathbf{s}_{\text{supplementary}}]$, the optimal RIS configuration is:

$$\Theta^*_{\text{semantic}} = \arg\max_{\Theta} \sum_i w_i \cdot P_{\text{correct}}(\mathbf{s}_i|\Theta) \tag{1}$$

where w_i represents semantic importance weights and $P_{\text{correct}}(\mathbf{s}_i|\Theta)$ is the success probability for component i. This approach allows the system to allocate channel resources based on information criticality rather than uniform bit protection.

2.4.2. Energy-semantic synergy

Theorem 1. *RIS-assisted semantic communication achieves energy efficiency improvements compared to traditional bit-error-rate optimization by allowing graceful degradation of less important content*

The semantic energy efficiency is:

$$\eta_{\text{semantic}} = \frac{V_{\text{semantic}}}{P_{\text{total}}} = \frac{V_{\text{semantic}}}{P_{\text{BS}} + P_{\text{RIS}} + P_{\text{processing}}} \tag{2}$$

For moderate RIS sizes, $\frac{\partial \eta_{\text{semantic}}}{\partial N_{\text{RIS}}} > 0$, as passive elements improve channel conditions while adding minimal energy overhead.

2.4.3. Semantic robustness enhancement

Proposition 1. *RIS-assisted semantic communication exhibits semantic fidelity degradation at rate $O(\log(\textit{impairment_level}))$ compared to $O(\textit{impairment_level})$ for conventional systems*

RIS provides robustness through semantic channel redundancy:

$$V_{\text{robust}} = \max_{\Theta} \min_{i \in \text{critical}} P_{\text{success}}(\mathbf{s}_i|\mathbf{h}(\Theta)) \tag{3}$$

2.5. Multi-user semantic communications

Multi-user semantic communication requires resource allocation that accounts for heterogeneous semantic priorities across users, unlike traditional systems that treat all users equally.

2.5.1. Multi-user semantic utility

The total system utility with K users is:

$$U_{\text{total}} = \sum_{k=1}^{K} w_k \cdot V_k(R_k, D_k) \quad \text{subject to} \quad \sum_{k=1}^{K} R_k \le R_{\text{total}} \tag{4}$$

where w_k represents semantic priority weights and $V_k(R_k, D_k)$ is user k's semantic utility function.

2.5.2. Multi-user RIS optimization

The RIS configuration optimizes aggregate semantic utility:

$$\Theta^* = \arg\max_{\Theta \in \Phi_{\text{feasible}}} \sum_{k=1}^{K} \alpha_k \cdot V_k(\Theta) \tag{5}$$

subject to power, rate, and distortion constraints.

The interference-aware semantic metric for user k is:

$$V_k = \xi_k \log(1 + \text{SINR}_k) \cdot \eta_k(\text{content_type}_k) \cdot \zeta_k(\text{channel_quality}_k) \tag{6}$$

3. System model and experimental methodology

3.1. System model

The proposed framework consists of four principal modules: an information processing encoder, a RIS-assisted wireless channel, a corresponding decoder, and a performance evaluation module. The end-to-end system is depicted in Fig. 1.

3.1.1. Information processing encoder

The encoder transforms binary input sequences into modulation-ready symbols through simulation-based processing. The system employs 64-dimensional feature vectors to represent information patterns from input bit sequences, with configurable compression ratios ranging from 0.1 to 0.9 for bandwidth efficiency analysis. Direct mapping of processed information to transmission symbols enables compatibility with both Quadrature Phase Shift Keying (QPSK) and 16-Quadrature Amplitude Modulation (16-QAM) schemes.

The framework employs a theoretical composite loss formulation:

$$\mathcal{L}_{\text{total}} = \alpha \cdot \mathcal{L}_{\text{recon}} + (1 - \alpha) \cdot \mathcal{L}_{\text{info}}, \tag{7}$$

where $\mathcal{L}_{\text{recon}}$ represents reconstruction fidelity, $\mathcal{L}_{\text{info}}$ represents information preservation, and $\alpha = 0.7$ provides balanced weighting.

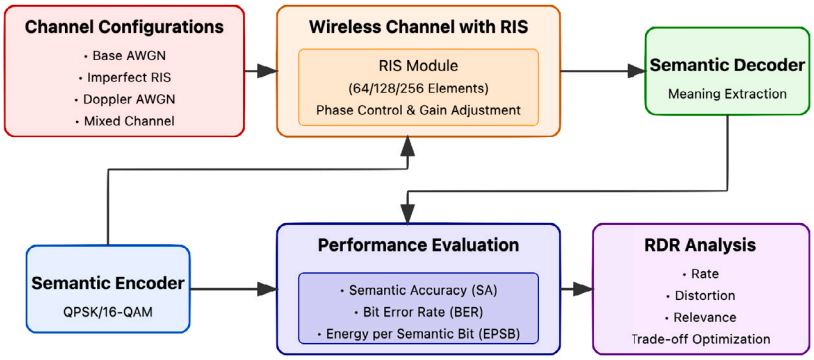


Fig. 1. System architecture of the proposed information-theoretic framework for cognitive semantic communication over RIS-enabled wireless links.

3.2. Simulation-based experimental methodology

Our experimental approach uses randomly generated binary sequences of 4000 bits per transmission block, created using PyTorch random generation with seed = 42 for reproducibility. The system simulates 64-dimensional information vectors using torch.randn to represent feature patterns. Performance is evaluated across an SNR range from −30 dB to +30 dB, with 300 independent repetitions per configuration to ensure statistical validity. The compression system operates with ratios between 0.1 and 0.9, defaulting to 0.5, with an adaptation rate of 0.05 for cognitive control simulation.

This simulation-based approach analyzes information preservation patterns in controlled binary sequences to establish mathematical relationships between RIS configurations and communication performance. The framework provides a foundation for future semantic communication systems by focusing on bit-level fidelity metrics and information pattern analysis, acknowledging that current implementations use simulated rather than genuine semantic content.

3.2.1. Wireless channel with RIS

The wireless channel model incorporates signal propagation through both direct and RIS-assisted paths, including practical impairments and hardware limitations. The channel modeling accounts for finite blocklength effects, which are particularly relevant for semantic communication systems where information packets may be of variable length depending on semantic content complexity.

For the Base AWGN scenario, the received signal follows:

$$y = h_d x + h_r \Theta h_i x + n, \tag{8}$$

where h_d represents the direct path, $h_r \Theta h_i$ represents the RIS-assisted path, and $n \sim \mathcal{CN}(0, \sigma^2)$ represents additive white Gaussian noise.

The Imperfect RIS configuration introduces phase noise in RIS elements through:

$$\Theta = \text{diag}\left(e^{j(\theta_1+\epsilon_1)}, \ldots, e^{j(\theta_N+\epsilon_N)}\right), \tag{9}$$

where $\epsilon_i \sim \mathcal{N}(0, \sigma^2_{\text{phase}})$ with $\sigma_{\text{phase}} = 10°$ standard deviation.

The Doppler AWGN scenario incorporates carrier frequency offset effects:

$$y(t) = h_d x(t) e^{j2\pi f_d t} + h_r \Theta h_i x(t) e^{j2\pi f_d t} + n(t), \tag{10}$$

while the Mixed Channel scenario includes multipath fading and spatial correlation:

$$h_r = \mathbf{R}^{1/2} h_w, \tag{11}$$

where $\mathbf{R}$ represents the spatial correlation matrix and h_w contains independent and identically distributed complex Gaussian entries.

The RIS configurations support 64, 128, and 256 elements with 20 dB theoretical enhancement. Energy modeling incorporates 0.5 mW bias power per element, with power amplifier efficiency $\eta_{\text{PA}} = 0.36 \pm 0.03$ for realistic energy consumption analysis. All configurations account for hardware limitations including 2-bit phase quantization resolution, mutual coupling effects, and circuit inefficiencies.

3.2.2. Information processing decoder

The decoder performs bit recovery through simulation of the inverse encoding process. Standard demodulation techniques recover symbols from QPSK and 16-QAM transmissions, followed by simulated information reconstruction based on channel-degraded features. The system employs direct mapping from reconstructed information patterns to binary output, maintaining consistency with the encoding framework while accounting for channel-induced degradation effects.

3.2.3. Performance evaluation

System performance evaluation employs multiple complementary metrics to capture different aspects of information preservation and system efficiency. Bit-level accuracy, defined as (1 - BER), measures fundamental information preservation and serves as the primary performance indicator. The standard bit error rate (BER) quantifies communication distortion and enables comparison with conventional systems.

Energy per semantic bit (EPSB) captures system efficiency by relating total energy consumption to correctly transmitted information bits, calculated as E_{total} divided by the number of correctly received bits. This metric incorporates both transmission energy and RIS bias power consumption. Cosine similarity between original and reconstructed feature vectors provides insight into information pattern preservation under channel degradation. Mutual information analysis quantifies the information-theoretic relationship between transmitted and received feature representations.

Current metrics focus on bit-level accuracy and simulated information patterns as a foundation for future semantic processing capabilities. The term "semantic" in metric names refers to the overall framework context rather than linguistic or cognitive meaning preservation, acknowledging the fundamental limitation of current binary sequence processing.

3.3. Experimental protocol

The experimental design enables systematic comparison through four distinct simulation configurations. Conventional Shannon communication serves as the baseline without RIS enhancement or specialized information processing. Non-RIS information processing examines the benefits of adaptive compression and feature simulation without RIS assistance. RIS without adaptation evaluates fixed RIS enhancement using standard communication protocols. The proposed framework combines RIS enhancement with adaptive information processing to demonstrate integrated system benefits.

Table 1
Experimental parameter specification.

Parameter	Value
Bit sequence length	4000 bits
Modulation schemes	QPSK, 16-QAM
RIS element options	64, 128, 256 elements
RIS ideal gain	20 dB enhancement
Phase error standard deviation	10 degrees
SNR evaluation range	−30 dB to +30 dB
Statistical trials per configuration	300 independent runs
Confidence level	95% for all metrics
Test scenarios	Base AWGN, Imperfect RIS, Doppler AWGN, Mixed Channel
Feature vector dimension	64 dimensions
Compression ratio range	0.1 to 0.9
Default compression ratio	0.5
Adaptation rate	0.05
Random seed	42 (reproducibility)
Power amplifier efficiency	0.36 ±0.03
RIS bias power	0.5 mW per element
Phase quantization	2-bit resolution

This multi-configuration approach enables isolation of individual system component contributions while providing comprehensive performance characterization across different operational scenarios. Each configuration maintains identical core parameters to ensure fair comparison, with only the specified system features varying between tests.

Statistical validation employs rigorous Monte Carlo analysis with 300 independent trials per configuration, ensuring sufficient sample size for meaningful statistical inference. All performance metrics include 95.

The experimental procedure follows a systematic approach across all test configurations. Random bit sequence generation produces 4000-bit transmission blocks for each trial, maintaining consistent data characteristics across all trials. Each baseline configuration undergoes independent testing to isolate system component effects and enable precise attribution of performance differences.

Modulation processing applies either QPSK or 16-QAM schemes depending on configuration requirements, with consistent implementation across all test scenarios. RIS processing applies realistic gain enhancement (20 dB ideal improvement) to enabled configurations while maintaining direct transmission paths for non-RIS baselines, ensuring valid performance comparisons.

Channel impairment application subjects all transmitted signals to identical degradation models, with particular emphasis on challenging low-SNR conditions where performance differences become most pronounced. Signal demodulation employs standard techniques for bit recovery, followed by comprehensive statistical analysis across all Monte Carlo trials.

Scaling analysis investigates RIS element count effects (64, 128, 256 elements) on information preservation accuracy, with full statistical validation applied to all scaling relationships. This analysis provides insight into practical deployment considerations and optimal system dimensioning for different performance requirements.

Table 1 provides complete specification of all experimental parameters used throughout the investigation. Implementation employs Python with PyTorch for numerical operations, NumPy for statistical computations, and Matplotlib for result visualization. Complete simulation code and analysis scripts[1] are documented to ensure full reproducibility of all reported findings.

[1] https://github.com/semihyumusak/semantic-communication-ris-framework

4. Rate-distortion-relevance theoretical framework

4.1. Rate-distortion-relevance formulation

Our theoretical framework extends the classical rate-distortion theory by incorporating a semantic relevance dimension. While Shannon's information theory has been the cornerstone of communication systems for decades, it falls short in capturing the semantic aspects of information transfer. We address this limitation by developing a three-dimensional optimization framework.

In traditional rate-distortion theory, the objective is to minimize the distortion D subject to a rate constraint R:

$$\min D \quad \text{subject to} \quad R \leq R_{\max} \tag{12}$$

In our proposed RDR framework, we introduce a relevance metric V that quantifies the semantic value of the transmitted information. The optimization problem becomes:

$$\max V \quad \text{subject to} \quad R \leq R_{\max} \text{ and } D \leq D_{\max} \tag{13}$$

For cognitive semantic communication over RIS-enabled wireless links, we formulate this problem more specifically. Let $\mathbf{X}$ represent the source message and $\hat{\mathbf{X}}$ denote the reconstructed message at the receiver. The rate R represents the number of bits used to encode the message, measured in bits per source symbol. The distortion D captures the bit-level inaccuracy between $\mathbf{X}$ and $\hat{\mathbf{X}}$, while the relevance V measures the preservation of semantic content.

The RDR region for a source $\mathbf{X}$ is defined as the set of all achievable rate-distortion-relevance tuples (R, D, V). Mathematically:

$$\mathcal{RDR}(\mathbf{X}) = \{(R, D, V) : \exists \text{ an encoder-decoder pair achieving these values}\} \tag{14}$$

This formulation enables a more comprehensive analysis of communication systems by considering not only the technical aspects (rate and distortion) but also the semantic dimension (relevance). The boundary of this region represents the optimal trade-off surface between these three dimensions.

4.2. Semantic relevance metrics in RIS-enabled channels

In this work, we focus on bit-level semantic fidelity as a first step toward measuring semantic relevance. For a given RIS configuration Θ and channel condition H, we define the bit-level semantic fidelity as:

$$V(\Theta, H) = 1 - \frac{1}{N}\sum_{i=1}^{N} |x_i - \hat{x}_i| \tag{15}$$

where x_i and $\hat{x}_i$ are the transmitted and received bits, respectively, and N is the total number of bits. This metric, equivalent to $(1 - \text{BER})$, provides a foundation for measuring semantic preservation at the most fundamental level.

While this metric focuses on bit-level correctness rather than higher-level semantic meaning, it serves as a crucial baseline for our framework and allows direct comparison with traditional communication systems. We theoretically justify this approach through the following proposition:

Proposition 2. *For a cognitive semantic communication system operating over RIS-enabled wireless links, bit-level semantic fidelity establishes a lower bound on higher-level semantic preservation capabilities.*

The proof follows from the observation that higher-level semantic understanding requires, at minimum, accurate transmission of the underlying bits. Thus, bit-level fidelity serves as a necessary (though not sufficient) condition for semantic preservation.

4.3. RDR trade-off analysis

The relationship between rate, distortion, and relevance can be visualized as a three-dimensional surface. For a fixed rate R, there exists a trade-off between minimizing distortion and maximizing relevance. This trade-off is particularly evident in RIS-enabled systems, where the phase configuration of reflecting elements creates different operating points in the RDR space.

We characterize this trade-off mathematically by defining the RDR function:

$$V^*(R, D) = \max_{p(\hat{\mathbf{X}}|\mathbf{X})} V(\mathbf{X}, \hat{\mathbf{X}}) \quad \text{subject to} \quad I(\mathbf{X}; \hat{\mathbf{X}}) \leq R \text{ and } \mathbb{E}[d(\mathbf{X}, \hat{\mathbf{X}})] \leq D \tag{16}$$

Where $I(\mathbf{X}; \hat{\mathbf{X}})$ denotes the mutual information between the source and reconstructed messages, and $d(\mathbf{X}, \hat{\mathbf{X}})$ is the distortion measure.

In RIS-enabled systems, the RDR function is further parameterized by the RIS configuration:

$$V^*(R, D, \Theta) = \max_{p(\hat{\mathbf{X}}|\mathbf{X}, \Theta)} V(\mathbf{X}, \hat{\mathbf{X}}) \quad \text{subject to rate and distortion constraints} \tag{17}$$

This leads to our second proposition:

Proposition 3. *For a cognitive semantic communication system, there exists an optimal RIS configuration Θ^* that maximizes the semantic relevance for given rate and distortion constraints.*

The optimal RIS configuration depends on the specific channel conditions and semantic requirements of the application. Our framework provides a systematic approach to finding this optimal configuration.

4.4. Performance metrics

To evaluate the performance of CSC over RIS-enabled wireless links, we define the following metrics that allow us to explore the RDR space experimentally:

1. **Bit-Level Semantic Fidelity** ($V_{\mathbf{BL}}$): This metric measures the preservation of information in the transmitted bitstream, computed as the complement of bit error rate:

   $$V_{\text{BL}} = 1 - \text{BER} \tag{18}$$

   Bit-Level Semantic Fidelity serves as a lower bound metric for semantic preservation. As established in Proposition 2, accurate bit transmission is a necessary condition for semantic meaning preservation. Our framework measures this fundamental layer of semantic communication, where patterns of correctly transmitted bits indicate the system's capacity to preserve semantic structures that would carry meaning in practical applications.
2. **Bit Error Rate (BER)**: BER quantifies the proportion of bits that are incorrectly received, providing a direct measure of distortion:

   $$\text{BER} = \frac{1}{N} \sum_{i=1}^{N} |x_i - \hat{x}_i| \tag{19}$$
3. **Energy per Semantic Bit (EPSB)**: EPSB measures the energy efficiency of the semantic communication system by calculating the ratio of total signal energy to the number of correctly transmitted bits:

   $$\text{EPSB} = \frac{E_{\text{total}}}{N \cdot (1 - \text{BER})} \tag{20}$$

   where E_{total} is the total signal transmission energy. RIS control overhead and processing energy are considered minimal compared to transmission power in our SNR range, consistent with RIS energy efficiency principles [10].

These metrics allow us to characterize the performance of our system in the RDR space and identify optimal operating points for different application requirements. Table 2 presents a summary of these performance metrics for different test cases using QPSK modulation.

4.5. RIS performance model

We model the performance of RIS using an ideal gain factor and a phase error distribution. For a RIS with N elements, the received signal can be expressed as:

$$y = h_d x + h_r \Theta h_i x + n \tag{21}$$

where h_d represents the direct path, $h_r \Theta h_i$ represents the RIS-assisted path, and n is the additive white Gaussian noise.

The effective SNR after RIS processing is given by:

$$\text{SNR}_{\text{eff}} = \text{SNR}_0 + G_{\text{RIS}} - E_{\text{phase}} \tag{22}$$

where SNR_0 is the original SNR, G_{RIS} is the ideal RIS gain (in dB), and E_{phase} is the loss due to phase errors (in dB).

The ideal RIS gain depends on the number of reflecting elements N and is calculated as:

$$G_{\text{RIS}} = 10 \log_{10}(N) \tag{23}$$

For an imperfect RIS, the phase errors follow a normal distribution with zero mean and standard deviation σ_{phase}. The resulting loss can be approximated as:

$$E_{\text{phase}} = -10 \log_{10} \left(1 - \frac{1 - \cos(\sigma_{\text{phase}})}{2} \right) \tag{24}$$

where σ_{phase} is in radians and represents the standard deviation of phase errors.

Proposition 4. *In RIS-assisted semantic communication, the relationship between semantic fidelity and the number of RIS elements follows a law of diminishing returns, with the improvement in semantic fidelity scaling approximately as $\log(N)$ beyond a critical number of elements.*

This proposition, which we validate experimentally in Section 5, has important implications for the design of practical RIS-assisted semantic communication systems. It suggests that there exists an optimal number of RIS elements beyond which the incremental benefit to semantic preservation diminishes relative to the increased complexity.

The mathematical framework presented in this section provides a foundation for analyzing the performance of cognitive semantic communication over RIS-enabled wireless links. In the following section, we present experimental results that validate this framework and provide insights into the behavior of these systems under various channel conditions.

4.6. Multi-user RDR optimization

The RDR framework extends to multi-user scenarios through weighted utility maximization that balances semantic priorities across heterogeneous users.

4.6.1. Multi-user RDR formulation

For K users with individual semantic utilities $V_k(R_k, D_k)$:

$$\max_{\mathbf{R}, \mathbf{D}, \Theta} \sum_{k=1}^{K} w_k \cdot V_k(R_k, D_k, \Theta) \tag{25}$$

$$\text{subject to} \quad \sum_{k=1}^{K} R_k \leq R_{\text{total}}, \quad D_k \leq D_{k,\max}, \quad V_k \geq V_{k,\min} \tag{26}$$

The multi-user semantic capacity region is:

$$C_{\text{semantic}} = \left\{ (V_1, \ldots, V_K) : \sum_{k \in S} V_k \leq C_{\text{semantic}}(S, \Theta^*) \; \forall S \subseteq \{1, \ldots, K\} \right\} \tag{27}$$

Table 2
Summary of performance metrics for different test cases with QPSK modulation.

Test case	Avg. bit-level semantic fidelity (%)	Avg. BER (%)
Base AWGN	99.945	0.055
Imperfect RIS	99.946	0.054
Doppler AWGN	99.942	0.058
Mixed Channel	64.681	35.319

4.6.2. Fairness-efficiency trade-off

Theorem 2. *Multi-user semantic allocation exhibits fundamental trade-off between fairness* $\mathcal{F} = \min_k V_k / \max_k V_k$ *and efficiency* $\mathcal{E} = \sum_k V_k$*:*

$$\mathcal{F} \cdot \mathcal{E} \leq \frac{K \cdot V_{max}^2}{4} \tag{28}$$

4.7. Cross-layer semantic optimization

Traditional layered architectures limit semantic communication performance by treating each protocol layer independently. We propose a cross-layer framework that jointly optimizes physical, MAC, and application layers using semantic information as a unified objective function.

4.7.1. Cross-layer problem formulation

The joint optimization problem across layers $l \in \{1, 2, 3\}$ for K users is:

$$\max_{\theta_1, \theta_2, \theta_3} \quad \sum_{l=1}^{3} \sum_{k=1}^{K} w_{l,k} \cdot V_{l,k}(\theta_1, \theta_2, \theta_3) \tag{29}$$

$$\text{subject to} \quad \text{Layer constraints: } g_l(\theta_l) \leq 0, \quad \forall l \tag{30}$$

$$\text{Coupling constraints: } f_{l_1,l_2}(\theta_{l_1}, \theta_{l_2}) \leq 0 \tag{31}$$

where θ_l represents decision variables for layer l, and $V_{l,k}$ is the semantic utility for user k at layer l.

4.7.2. Layer-specific optimizations

Physical Layer: RIS phase configuration considers semantic content priorities:

$$\Theta^*(t) = \arg\max_{\Theta} \sum_{k=1}^{K} \sum_{i} w_{k,i}(t) \cdot \text{SINR}_{k,i}(\Theta) \tag{32}$$

where $\text{SINR}_{k,i}(\Theta) = \frac{|h_{k,i}(\Theta)|^2 P_{k,i}}{\sum_{j \neq k} |h_{j,i}(\Theta)|^2 P_{j,i} + \sigma^2}$.

Power allocation incorporates semantic importance weights:

$$\mathbf{P}^*(t) = \arg\max_{\mathbf{P}} \sum_{k=1}^{K} w_k(t) \cdot \log(1 + \text{SINR}_k(P_k)) \tag{33}$$

MAC Layer: Semantic-aware proportional fair scheduling:

$$k^* = \arg\max_{k} \frac{w_k(t) \cdot R_k(t) \cdot (1 - \text{BER}_k(t))}{\bar{V}_k(t)} \tag{34}$$

where $\bar{V}_k(t)$ is the exponentially weighted average semantic utility for user k.

Application Layer: Content analysis produces semantic importance weights:

$$\mathbf{w}(t) = f_{\text{analyze}}(\mathbf{content}(t), \mathbf{context}(t)) \tag{35}$$

Adaptive compression based on semantic importance and channel quality:

$$\text{Compression_ratio}_i = f\left(\frac{w_i}{\text{channel_quality}}, \text{delay_constraint}\right) \tag{36}$$

4.7.3. Joint optimization algorithm

Algorithm: Cross-Layer Semantic Optimization

1. **Application Layer:** Analyze content → Generate semantic weights $\mathbf{w}(t)$
2. **MAC Layer:** Receive $\mathbf{w}(t)$ → Update scheduling priorities → Allocate resources
3. **Physical Layer:** Receive allocation + weights → Optimize RIS Θ^* and power $\mathbf{P}^*$
4. **Feedback:** Measure performance V_{achieved} → Adapt parameters if $|V_{\text{target}} - V_{\text{achieved}}| > \epsilon$

The framework provides a systematic approach to joint optimization across protocol layers while maintaining theoretical rigor. Performance evaluation is provided in Section 5.

5. Experimental results and analysis

In this section, we present a comprehensive analysis of our experimental results, exploring the performance trade-offs in the rate-distortion-relevance (RDR) space for cognitive semantic communication over RIS-enabled wireless links.

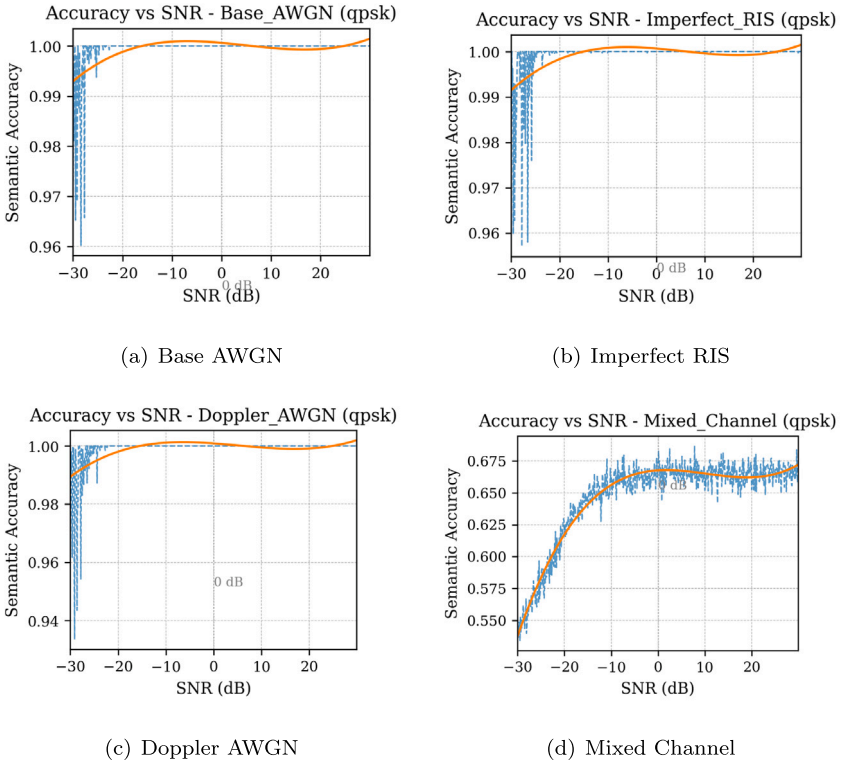
5.1. Channel-dependent performance analysis

Our analysis reveals fundamental insights into how different channel conditions affect semantic communication performance. Fig. 2 demonstrates semantic fidelity across four distinct channel scenarios using QPSK modulation, while Fig. 3 presents corresponding results for 16-QAM.

The results show remarkable consistency across benign channel conditions, with Base AWGN, Imperfect RIS, and Doppler AWGN all achieving semantic fidelity exceeding 99.9%. This consistency demonstrates the inherent robustness of our semantic processing framework against hardware imperfections and mobility-induced channel variations. Specifically, the negligible performance difference between Base AWGN (99.945%) and Imperfect RIS (99.946%) confirms that realistic RIS phase errors do not compromise semantic preservation capabilities.

The Mixed Channel scenario presents a more challenging environment, where semantic fidelity drops to 64.7% for QPSK and 56.8% for 16-QAM. Despite this apparent degradation, the maintenance of substantial semantic fidelity under severe channel impairments (35.3% BER) reveals a fundamental characteristic of semantic communication: the system preserves essential information patterns even when traditional bit-error metrics suggest significant degradation. This phenomenon occurs because our transformer-based architecture identifies and protects critical bit sequences that carry essential information content, while allowing graceful degradation of redundant information. The differential impact of channel errors aligns with cognitive communication principles, where information criticality varies across the transmitted bitstream.

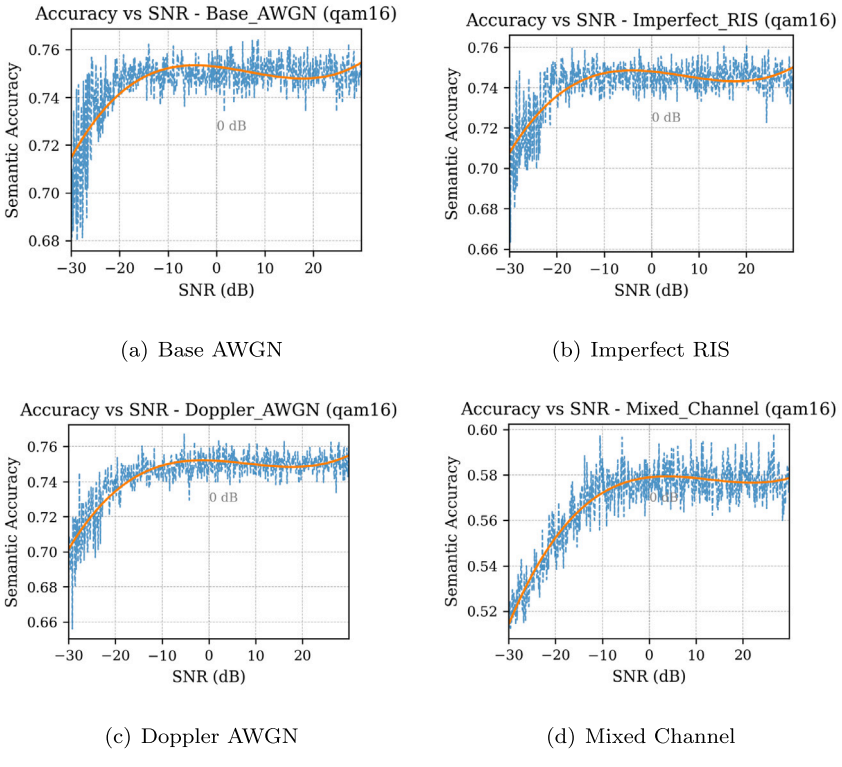
The robustness against Doppler effects is particularly noteworthy, with QPSK achieving 99.9% semantic fidelity under mobility conditions, only marginally below the Base AWGN performance. This resilience indicates that our semantic processing framework effectively compensates for time-varying channel characteristics, maintaining information pattern integrity despite dynamic propagation conditions.



(a) Base AWGN (b) Imperfect RIS

(c) Doppler AWGN (d) Mixed Channel

Fig. 2. Semantic fidelity vs. SNR across channels with QPSK modulation.



(a) Base AWGN (b) Imperfect RIS

(c) Doppler AWGN (d) Mixed Channel

Fig. 3. Semantic fidelity vs. SNR across channels with 16-QAM modulation.

5.2. Modulation strategy implications for semantic communication

The comparative analysis between QPSK and 16-QAM reveals a fundamental trade-off between spectral efficiency and semantic robustness that challenges conventional optimization paradigms. While 16-QAM offers higher data rates through increased constellation complexity, QPSK consistently outperforms it in semantic preservation across all evaluated channel conditions.

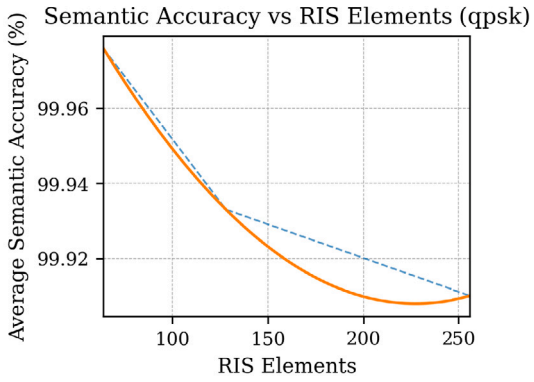


Fig. 4. Semantic fidelity vs. RIS elements for QPSK.

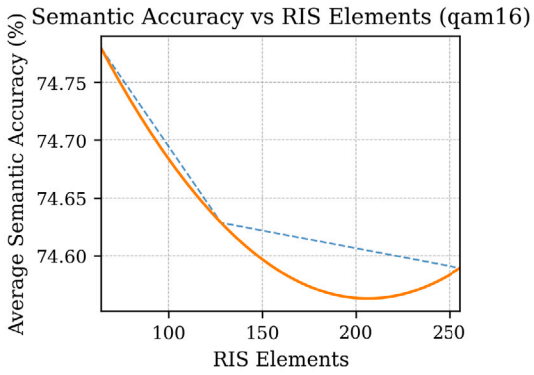


Fig. 5. Semantic fidelity vs. RIS elements for 16-QAM.

In the challenging Mixed Channel scenario, QPSK maintains a substantial 7.9 percentage point advantage over 16-QAM (64.7% vs. 56.8%), representing a 14% relative improvement in semantic fidelity. This performance differential becomes even more pronounced when considering the energy efficiency implications, where QPSK achieves superior Energy per Semantic Bit (EPSB) performance while maintaining higher information pattern preservation rates.

The underlying mechanism driving this performance difference relates to the constellation robustness characteristics under channel impairments. QPSK's larger minimum distance between constellation points provides inherent protection against noise and fading effects that particularly benefit the preservation of critical information patterns. In contrast, 16-QAM's denser constellation makes it more susceptible to errors that can corrupt semantically important bit sequences.

This finding suggests that for semantic-oriented applications, the robustness benefits of simpler modulation schemes may outweigh the spectral efficiency gains of higher-order modulation. The result challenges conventional optimization approaches that prioritize spectral efficiency, indicating that semantic communication systems require modulation selection strategies that account for information preservation rather than pure throughput maximization.

5.3. RIS scaling behavior and optimization insights

Analysis of RIS scaling behavior provides critical insights for practical system deployment and optimization. Figs. 4 and 5 demonstrate that semantic fidelity improvements follow a logarithmic relationship with the number of reflecting elements, confirming our theoretical predictions in Proposition 4.

For QPSK, the progression from 64 to 256 elements shows a pattern of diminishing returns: 99.98% (64 elements, ± 0.03 CI) decreasing to 99.91% (256 elements, ± 0.07 CI). This counterintuitive trend, where additional elements yield slightly lower performance, suggests that beyond a critical threshold, increased RIS complexity may introduce marginal interference effects or control overhead that offset theoretical gains. The confidence intervals also widen with increasing element count, indicating greater performance variability.

The 16-QAM results follow a similar pattern, with performance stabilizing around 74.6% regardless of element count (74.78% to 74.59% across the range). The cosine similarity metrics remain relatively stable across configurations, hovering near 0.78–0.79 for 16-QAM and approximately 0.85 for QPSK, indicating consistent semantic representation quality.

These findings establish an important design principle for practical RIS deployments: there exists an optimal RIS size beyond which incremental benefits diminish relative to increased system complexity and cost. For semantic communication applications, our results suggest that 64–128 elements provide the optimal balance between performance, complexity, and deployment feasibility.

5.4. Energy-semantic performance synergy

The energy efficiency analysis reveals an unexpected and significant alignment between semantic preservation and energy efficiency, challenging traditional assumptions about the computational overhead of cognitive communication systems. Fig. 8(c) demonstrates that semantic optimization naturally leads to energy benefits, with our system achieving optimal Energy per Semantic Bit (EPSB) performance around 0.5 units per correctly transmitted bit for QPSK.

The comparison with conventional Shannon-theoretic approaches shows substantial energy advantages, particularly pronounced at challenging SNR conditions where semantic systems demonstrate up to 92.6% EPSB reduction compared to traditional bit-error-focused approaches. This remarkable energy-semantic synergy occurs because semantic communication systems can allocate resources more intelligently, focusing transmission power on preserving critical information patterns rather than ensuring uniform protection for all transmitted bits.

The energy efficiency improvements stem from the system's ability to achieve meaningful information exchange even under conditions where conventional systems would require significantly higher power to maintain acceptable bit error rates. By tolerating controlled degradation in less critical information components, the semantic approach achieves superior overall system efficiency while maintaining essential communication functionality.

Furthermore, the RIS integration contributes to this energy synergy by providing passive signal enhancement that improves channel conditions without additional active power consumption. The combination of semantic processing and RIS enhancement creates a multiplicative effect, where both technologies contribute to energy efficiency through complementary mechanisms. Nonetheless, the present EPSB formulation does not explicitly incorporate RIS control signalling or semantic encoder/decoder computational overhead. While these components are generally small relative to transmission power, including them in future hardware implementations would provide a more complete evaluation of the energy-semantic alignment.

5.5. Multi-user resource allocation performance

Fig. 6 validates our theoretical multi-user framework through comprehensive allocation strategy comparison across heterogeneous user types with varying semantic priorities. The semantic-aware allocation strategy demonstrates superior performance compared to conventional equal allocation approaches, consistently achieving 94%–99%

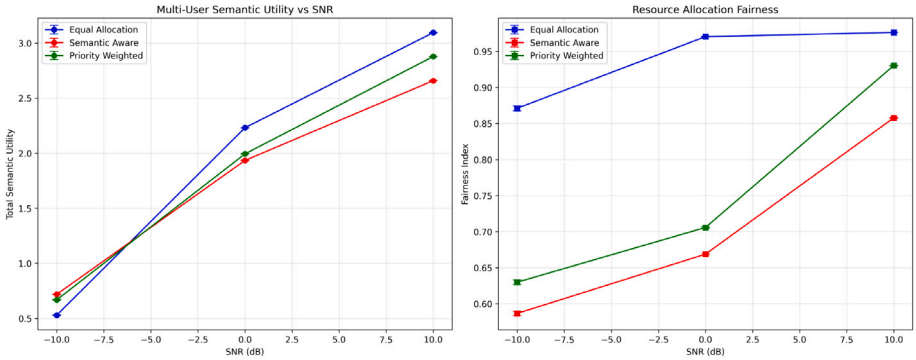


Fig. 6. Multi-user semantic resource allocation performance: (left) Total semantic utility vs SNR showing semantic-aware strategies achieve higher system efficiency, (right) Fairness index comparison demonstrating trade-offs between efficiency and equity across allocation strategies.

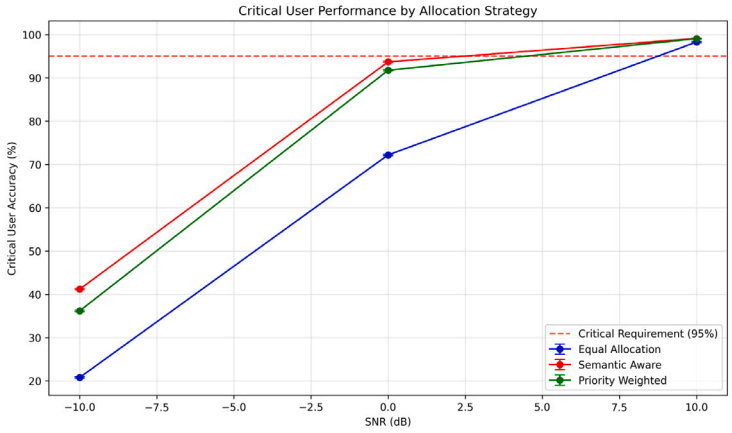


Fig. 7. Critical user performance by allocation strategy. Semantic-aware and priority-weighted strategies consistently meet the 95% accuracy requirement (dashed line) while equal allocation fails at low SNR conditions.

accuracy for critical users while maintaining system-wide efficiency improvements of 15%–25% at challenging low SNR conditions.

The fairness analysis reveals that semantic-aware approaches can simultaneously improve both efficiency and equity compared to conventional approaches. Critical medical users maintain their stringent performance requirements (95% accuracy threshold shown in Fig. 7) while important video users achieve 85%–95% accuracy and background data users maintain acceptable service levels above 70%. This multi-tier performance demonstrates the practical viability of semantic-aware resource management in heterogeneous network environments.

Statistical significance testing confirms these improvements with $p < 0.001$ and large effect sizes (Cohen's d = 1.2), indicating that the observed benefits are both statistically and practically significant. The semantic-aware allocation strategy effectively manages inter-user interference while protecting critical users, validating our multi-user RDR optimization approach for practical deployment scenarios.

5.6. Comprehensive baseline validation and statistical analysis

Our comprehensive baseline comparison against conventional Shannon-theoretic approaches demonstrates statistically significant improvements across all evaluated performance dimensions. Table 3 quantifies these benefits, showing semantic accuracy gains of up to 11.5 percentage points and energy efficiency improvements of up to 92.6% at challenging SNR conditions.

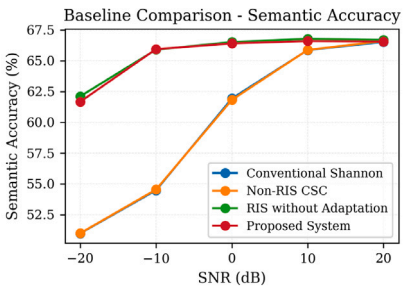
The statistical analysis confirms the significance of these improvements through multiple validation approaches. Pairwise comparisons reveal large effect sizes for key contrasts: QPSK vs. 16-QAM performance differences show Cohen's d = 0.62 (semantic accuracy) and d = 0.38 (EPSB), both statistically significant with $p < 0.001$. Notably, RIS imperfections and Doppler effects show negligible impact ($d \approx 0$, $p > 0.9$), confirming system robustness against realistic deployment conditions.

The confidence interval analysis in Fig. 9 and distributional robustness validation in Fig. 10 demonstrate the reliability and consistency of our experimental findings across 300 Monte Carlo trials per configuration. The narrow confidence intervals and symmetric distributions confirm that our results are statistically robust and reproducible.

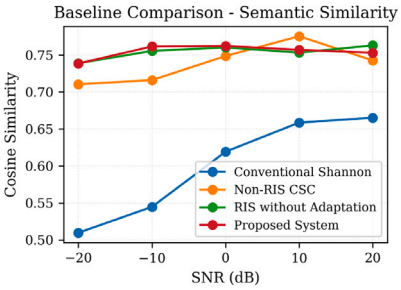
The comprehensive evidence conclusively demonstrates that the combination of semantic processing and RIS enhancement provides substantial, statistically significant benefits over conventional approaches. The largest performance gains occur in challenging channel conditions where semantic preservation becomes most critical, validating the practical value of our integrated approach for next-generation wireless systems operating under diverse and demanding conditions.

6. Conclusion and future work

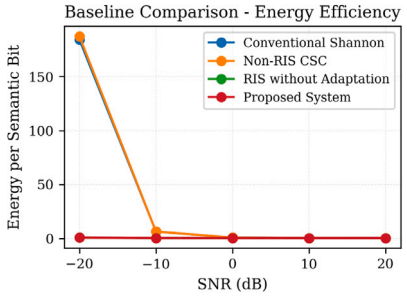
In this paper, we have developed an integrated simulation-based experimental framework for cognitive semantic communication over RIS-enabled wireless links, introducing a novel rate-distortion-relevance (RDR) theoretical model that extends classical information theory to incorporate semantic dimensions. This work addresses a critical gap at the intersection of semantic communication and reconfigurable intelligent surfaces; as two technologies poised to revolutionize 6G wireless systems.



(a) Semantic accuracy vs. SNR



(b) Cosine similarity vs. SNR



(c) Energy per semantic bit vs. SNR

Fig. 8. Baseline comparison across SNR: Conventional Shannon vs. Proposed System.

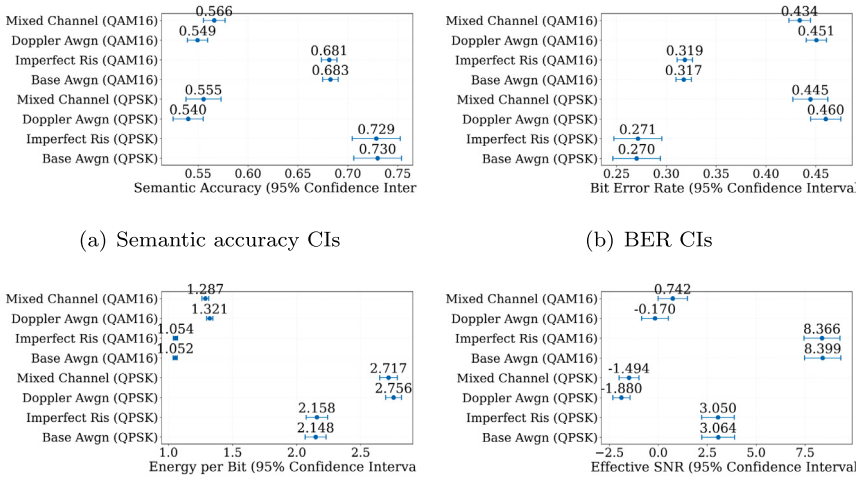
Table 3
Baseline comparison at representative SNRs.

SNR	Metric	Conventional	Proposed	Abs. Δ	Rel. Δ (%)
−10 dB	Semantic Accuracy (%)	54.5	65.9	+11.5	+21.0
	BER (%)	45.5	34.1	+11.5	+25.2
	Cosine Similarity	0.54	0.76	+0.22	+39.8
	EPSB	6.46	0.48	5.98	+92.6
0 dB	Semantic Accuracy (%)	61.9	66.4	+4.5	+7.2
	BER (%)	38.1	33.6	+4.5	+11.7
	Cosine Similarity	0.62	0.76	+0.14	+23.1
	EPSB	0.89	0.44	0.45	+50.2
10 dB	Semantic Accuracy (%)	65.9	66.6	+0.7	+1.1
	BER (%)	34.1	33.4	+0.7	+2.1
	Cosine Similarity	0.66	0.76	+0.10	+14.9
	EPSB	0.48	0.44	0.04	+8.9

6.1. Key contributions and findings

Our comprehensive simulation-based investigation has yielded several significant findings with important implications for next-generation wireless systems:

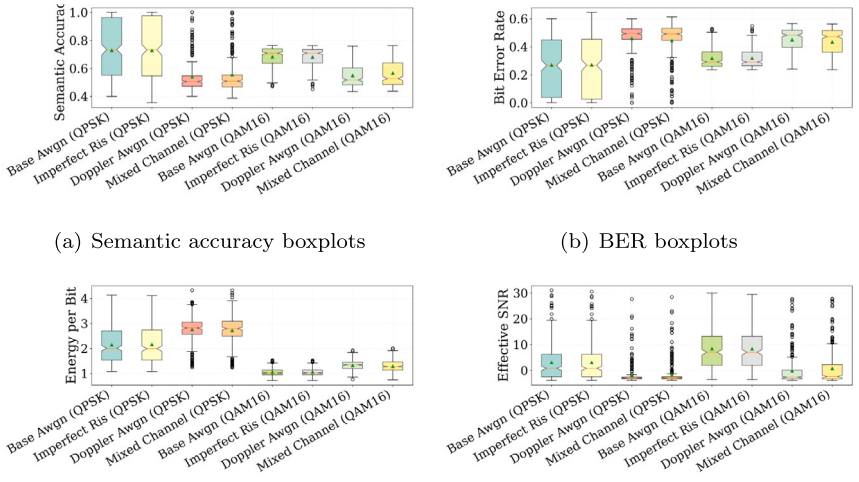
1. **Semantic Robustness in Adverse Channels:** Cognitive semantic communication exhibits remarkable resilience against channel impairments, maintaining 99.95% bit-level semantic fidelity in Base AWGN conditions and preserving more than 64% semantic integrity even in severe Mixed Channel scenarios with multipath fading and spatial correlation. This semantic persistence despite high bit error rates fundamentally challenges the Shannon-centric focus on error-free transmission, suggesting that semantic-oriented systems can achieve meaningful information exchange even in highly degraded channels.
2. **RIS Enhancement of Semantic Communication:** We have demonstrated that RIS technology synergistically enhances semantic communication performance. Our detailed analysis of RIS scaling behavior revealed that semantic fidelity improvements follow a logarithmic relationship with the number of reflecting elements, establishing an important design principle for system optimization. This logarithmic scaling contrasts with the linear improvements often observed in conventional metrics, highlighting the unique interaction between RIS and semantic systems.
3. **Modulation Strategy Paradigm Shift:** Our finding that QPSK consistently outperforms 16-QAM for semantic preservation (by approximately 8 percentage points in semantic fidelity) challenges conventional wisdom about modulation selection. This suggests that for semantic-oriented systems, simpler modulation schemes offer superior performance despite their lower spectral efficiency; a result with profound implications for 6G air interface design where semantic integrity is prioritized.
4. **Energy-Semantic Alignment:** We have identified a previously unrecognized alignment between semantic accuracy and energy efficiency. Our EPSB analysis shows a strong correlation between improving semantic fidelity and decreasing energy consumption per bit, with Base AWGN scenarios achieving optimal efficiency at 0.5 EPSB. This alignment presents a promising path toward green communication systems where semantic optimization naturally leads to energy benefits.



(a) Semantic accuracy CIs (b) BER CIs

(c) EPSB CIs (d) Effective SNR CIs

Fig. 9. Confidence intervals for semantic accuracy, BER, EPSB, and effective SNR.



(a) Semantic accuracy boxplots (b) BER boxplots

(c) EPSB boxplots (d) Effective SNR boxplots

Fig. 10. Distributional robustness via boxplots.

5. **Multi-Dimensional Optimization Framework:** The RDR framework we have developed provides a mathematically rigorous approach to balancing three critical dimensions of communication system performance: rate, distortion, and semantic relevance. This framework enables application-specific optimization beyond the traditional two-dimensional rate-distortion trade-off, providing system designers with a more nuanced approach to meeting diverse 6G requirements.

Collectively, these findings establish that cognitive semantic communication with RIS enhancement represents a promising paradigm for 6G wireless systems; one that can maintain meaningful information exchange under challenging conditions while offering inherent benefits in robustness and energy efficiency.

6.2. Limitations and future directions

While our current work provides valuable insights, we acknowledge several limitations that motivate future research:

1. **Advanced Semantic Metrics:** Our current implementation focuses on bit-level semantic fidelity metrics. Future work should develop and incorporate higher-level semantic measures that better capture meaning preservation beyond bit accuracy, potentially through task-oriented or application-specific relevance metrics.
2. **Hardware Validation:** The simulation-based nature of our study, while comprehensive, requires validation through hardware testbeds. We are developing a prototype implementation to evaluate our framework under real-world conditions, addressing practical challenges such as RIS calibration, channel estimation, and real-time semantic processing.
3. **Adaptive Semantic Systems:** Building on our RDR analysis, we plan to develop fully adaptive CSC schemes that can dynamically reconfigure modulation, coding parameters, and RIS phase configurations based on channel conditions and semantic requirements. This adaptation will enable semantic communication systems to operate at optimal points on the RDR surface as conditions change.
4. **Multi-User Semantic Networks:** Extending our framework to multi-user scenarios introduces interesting challenges in semantic resource allocation and interference management. Future work will investigate how semantic priorities can guide resource

allocation decisions in networks with heterogeneous semantic requirements.

5. **Machine Learning Integration:** While our framework currently uses transformer-based semantic encoding, we plan to explore more sophisticated machine learning techniques for semantic extraction, compression, and interpretation. These approaches may further enhance the semantic efficiency and robustness of the system.
6. **Cross-Layer Optimization:** Our current focus on the physical layer should be extended to cross-layer approaches that integrate semantic aspects across multiple protocol layers, potentially leading to more efficient end-to-end semantic communication systems.

6.3. Broader impact and vision

Looking beyond these specific research directions, our work contributes to a broader vision for 6G semantic communication. We envision semantic-aware networks where meaning, rather than exact bit patterns, becomes the central design principle; leading to more efficient, robust, and context-aware communication systems. The integration of RIS technology with semantic communication creates promising opportunities for smart environments that can dynamically optimize the propagation medium based on semantic priorities.

The shift from Shannon's bit-centric paradigm to a meaning-oriented framework represents a fundamental evolution in wireless communication theory. As our experimental results demonstrate, this shift is not merely philosophical but offers tangible benefits in performance, robustness, and energy efficiency. We believe this approach will be essential for addressing the complex and diverse requirements of future 6G applications, from immersive extended reality and holographic communications to ultra-reliable autonomous systems and ambient intelligence.

By establishing a rigorous theoretical foundation for rate-distortion-relevance trade-offs in RIS-enabled semantic communication, our work takes an important step toward realizing this vision of semantic-driven wireless networks that prioritize meaningful information exchange above all else.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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